

UNDERSTANDING MULTI-TEACHER ON-POLICY DISTILLATION: FROM GRADIENTS TO CAPABILITIES

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ABSTRACT

Multi-teacher on-policy distillation (MOPD) combines domain specialists into a single student, but how loss normalization, optimizer dynamics, and vocabulary supervision shape its updates and task performance remains unclear. We study these choices with Qwen3-1.7B and four reinforcement-learned teachers. First, token- and response-level averaging produce distinct gradients but similar final scores under top-64 supervision and Adam; the spread is larger under sampled-token supervision with SGD, while sampled-token Adam endpoints are nearly tied. Second, joint training changes about 97% of FP32 parameters, whereas only 7.3–8.5% differ after BF16 rounding. The apparent alignment of teacher-specific updates largely reflects Adam’s accumulated moments: subtracting the zero-gradient step reduces their mean cosine from above 0.83 to below 0.01 on domain-specific inputs. Third, the top-64 intersection retains over 99.9% of student probability on average at the evaluated Qwen checkpoints, yet provides no consistent task advantage over sampled-token supervision. SmolLM3-3B diagnostics further reveal low-coverage tails and appreciable gradient error despite high mean coverage and cosine. These results distinguish current gradients from optimizer history and stored-weight changes, and show why vocabulary supervision should be assessed through per-task learning curves alongside parameter statistics.

1 INTRODUCTION

Multi-teacher on-policy distillation (MOPD) transfers specialized capabilities into a single model (Xiaomi LLM-Core Team, 2026; NVIDIA, 2026; Kimi Team et al., 2026; DeepSeek-AI et al., 2026). The student generates responses, and a teacher assigned to each prompt domain supplies token-level supervision. Improving several capabilities requires deciding how much each domain and response contributes to the loss, and how many next-token alternatives the teacher supervises.

Recent work establishes both the potential of MOPD and its sensitivity to these choices. Ma et al. (2026) study capability integration with policy-gradient and top- k objectives. Open-MOPD addresses uneven transfer through domain token balancing, performance-gap weighting, and refreshed rewards (Gao et al., 2026). The Nemotron 3 Ultra report finds no advantage for top- k or full-vocabulary matching over sampled-token supervision in preliminary experiments (NVIDIA, 2026, Section 3.3.5). Other studies report parameter changes concentrated in a small set of stored weights (Yu et al., 2026; Shen et al., 2026). How these observations connect loss construction, optimizer updates, and task performance remains open.

We distinguish three effects. Equal prompt counts do not imply equal domain loss weights when response lengths differ; fixing domain weights still leaves a choice between token and response averaging. Adam mixes the current gradient with accumulated moments, while BF16 storage can hide small FP32 changes. Finally, supervising more vocabulary tokens need not improve every task. We examine these effects with Qwen3-1.7B and four domain teachers:

1. **Normalization changes gradients more than final top-64/Adam scores (Section 3).** Domain-token and domain-response gradients have mean cosine 0.680, rising to 0.960 after applying the same Adam state. The three averaging rules finish within 0.17 mean-score points, although their learning curves differ.

2. **BF16 sparsity and Adam alignment obscure current teacher signals (Section 4).** Joint training changes about 97% of FP32 weights but only 7.3–8.5% of BF16 weights. Subtracting Adam’s zero-gradient step reduces the mean teacher-update cosine from above 0.83 to below 0.01 on domain-specific inputs.
3. **More vocabulary supervision does not consistently improve task performance (Section 5).** Top-64 and full-vocabulary supervision yield similar single-step concentration. Against sampled-token supervision, top-64’s joint-training mean advantage shrinks from 1.69 points at update 250 to 0.43 at update 500, with mixed per-task differences.

SmolLM3-3B diagnostics extend the vocabulary comparison to lower-coverage conditions: high average intersection mass can conceal a low-coverage tail, and high gradient cosine can coexist with appreciable vector error. Section 6 connects these measurements to checkpoint selection: a per-domain retention constraint leaves all ten development-selected checkpoints unchanged. Together, the results motivate per-task learning curves alongside gradient and parameter statistics.

2 PRELIMINARIES

Multi-teacher on-policy distillation (MOPD). For domain d , the student generates $y \sim p_\theta(\cdot | x)$ for $x \sim \mathcal{D}_d$, and a frozen teacher q_d supplies probabilities at each prefix $h_t = (x, y_{<t})$ (Gao et al., 2026). For a rollout batch $\mathcal{B}$, response i has T_i valid tokens and mean loss $\ell_i = T_i^{-1} \sum_t \ell_{i,t}$. The objective is

$$\mathcal{L}(\theta) = \sum_{i \in \mathcal{B}} w_i \ell_i(\theta), \quad w_i \geq 0, \quad \sum_{i \in \mathcal{B}} w_i = 1. \quad (1)$$

The token loss determines the vocabulary supervision; the response weights w_i determine normalization across tokens, responses, and domains. Each update uses fresh student responses, held fixed with their masks and weights during gradient computation.

Sampled-token, top- k , and full-vocabulary supervision. Write $p_\theta(v) = p_\theta(v | h_t)$ and $q_d(v) = q_d(v | h_t)$. Full-vocabulary supervision minimizes reverse KL over $\mathcal{V}$; the policy-gradient (PG) loss scores the sampled token y_t :

$$\ell_{\text{full}} = \sum_{v \in \mathcal{V}} p_\theta(v) \log \frac{p_\theta(v)}{q_d(v)}, \quad \ell_{\text{PG}}(y_t) = -\text{sg}[\log q_d(y_t) - \log p_\theta(a)] \log p_\theta(y_t) \quad (2)$$

Here sg denotes stop-gradient. The expected PG gradient equals the full reverse-KL gradient: differentiating Equation 2 gives $\mathbb{E}_{a \sim p_\theta} [\log(p_\theta(a)/q_d(a)) \nabla_\theta \log p_\theta(a)]$, since $\sum_v \nabla_\theta p_\theta(v) = 0$.

Top- k supervision uses $I = S \cap T$, where S, T are the student and teacher top- k sets, computed at rollout and held fixed. Its loss is

$$\ell_{\text{top-}k} = - \sum_{v \in I} \text{sg} \left[\frac{p_\theta(v)}{\sum_{u \in S} p_\theta(u)} (\log q_d(v) - \log p_\theta(v)) \right] \log p_\theta(v). \quad (3)$$

Probabilities use the full-vocabulary softmax; detached weights are normalized over S , and an empty intersection gives zero loss. I16 and I64 denote $k = 16$ and 64. We call the number of supervised vocabulary tokens the *supervision density*: one for PG, $|I| \leq k$ for intersection supervision, and $|\mathcal{V}|$ for full supervision.

Domain weighting and loss normalization. Let $\mathcal{B}_d$ contain responses from domain d , and let $\lambda_d \geq 0$ with $\sum_d \lambda_d = 1$ be the target domain loss weights. For nonempty responses and domains, the three rules assign response $i \in \mathcal{B}_d$ weights

$$w_i^{\text{GT}} = \frac{T_i}{\sum_{j \in \mathcal{B}} T_j}, \quad w_i^{\text{DT}} = \frac{\lambda_d T_i}{\sum_{j \in \mathcal{B}_d} T_j}, \quad w_i^{\text{DR}} = \frac{\lambda_d}{|\mathcal{B}_d|}. \quad (4)$$

Global-token (GT) averaging weights domains by their response-token shares. Domain-token (DT) and domain-response (DR) averaging fix each domain’s total weight to λ_d : DT weights responses by length, while DR weights their mean losses equally. Domain loss weights are distinct from prompt sampling proportions.

Update similarity. Let u_d be the parameter change from one step on domain d 's mean loss. Each compared step starts from identical student weights and optimizer state, including Adam's moments. For nonzero updates, their cosine is $C_{de} = (u_d^\top u_e) / (\|u_d\| \|u_e\|)$.

3 LOSS NORMALIZATION AND TASK PERFORMANCE

3.1 EQUAL DOMAIN WEIGHTS DO NOT REMOVE RESPONSE-LENGTH WEIGHTING

DT and DR both assign domain d weight λ_d , but only DT weights responses by length. Their gradient difference depends on how response length covaries with gradient direction.

For response gradients $g_i = \nabla_{\theta} \ell_i$ and domain means $\bar{g}_d, \bar{T}_d$, the token-weighted gradient is

$$\frac{\sum_{i \in \mathcal{B}_d} T_i g_i}{\sum_{i \in \mathcal{B}_d} T_i} = \bar{g}_d + \frac{\text{Cov}_d(T, g)}{\bar{T}_d}. \quad (5)$$

DT retains the length-gradient covariance term, DR uses $\bar{g}_d$ alone. Correcting each domain's GT contribution by its target weight divided by its observed token share gives DT exactly, as in OpenMOPD (Gao et al., 2026). This correction leaves within-domain length weighting intact.

3.2 SIMILAR I64 ENDPOINT SCORES DESPITE DIFFERENT LOSS WEIGHTS

Experimental setting. We train Qwen3-1.7B for 500 updates with four RL-specialized teachers derived from the same base model for mathematics, code, instruction following, and science. Each update uses 16 fresh responses per domain. The I64 GT/DT/DR runs share initialization, teachers, prompt schedule, and AdamW settings. We evaluate MATH-500, LiveCodeBench v6 (1,055 questions), IFBench, and GPQA Diamond, reporting their equally weighted mean. The first three use greedy accuracy; GPQA averages correctness over four sampled responses per question. Thinking is disabled throughout.

A threefold loss-weight imbalance yields close I64 point estimates. Mathematics and instruction following each receive 25% of prompts but account for 44.05% and 14.98% of GT's valid response tokens in the first 50 batches. DT and DR fix both loss weights at 25% (Figure 1a). Yet update-500 mean scores are 39.04% (DR), 39.19% (DT), and 39.02% (GT), spanning 0.17 percentage points. The preferred rule varies by task and checkpoint: all three approach teacher performance on mathematics and instruction following, retain a code gap, and start near the science teacher's score. DT exceeds GT by 0.17 mean points, with an estimated paired 95% interval of $[-1.27, 1.61]$ (Appendix A.3); the close point estimates do not establish equivalence.

Normalization has a larger observed score spread under PG/SGD. The endpoint score range across rules is 0.01 points for PG/Adam and 1.39 for PG/SGD, versus 0.17 for I64/Adam (Table 1). PG/SGD favors DR over GT by 1.39 points, with an estimated paired 95% interval of $[-0.04, 2.82]$; PG/Adam is nearly tied. The observed normalization spread depends on the training configuration, but these comparisons do not establish a difference between rules (Table 5). All ten configurations improve the initial mean from 33.94% to 38.58–39.96%.

3.3 ADAM REDUCES DIRECTIONAL DIFFERENCES BETWEEN NORMALIZATION RULES

Different gradients produce similar Adam updates. On six fixed batches at joint-PG checkpoints 100 and 250, I64 gradient cosines average 0.680 (DR–DT), 0.483 (DR–GT), and 0.822 (DT–GT). Student weights, responses, masks, and teacher scores are shared, isolating loss normalization. Applying the gradients to identical copies of the saved Adam moments raises FP32 update cosines to 0.960, 0.939, and 0.979. Adam thus produces much more aligned updates from directionally different gradients.

Clipping changes magnitude, not direction. Global norm clipping preserves pairwise gradient cosines. DR is clipped on all 500 updates and retains a median 9.3% of the gradient norm; DT and GT are clipped on 46.2% and 15.0% of updates (Figure 2a). Clipping cannot explain Adam's higher update cosine. Resetting its moments also changes update norms (panel b), linking update magnitude to optimizer history.

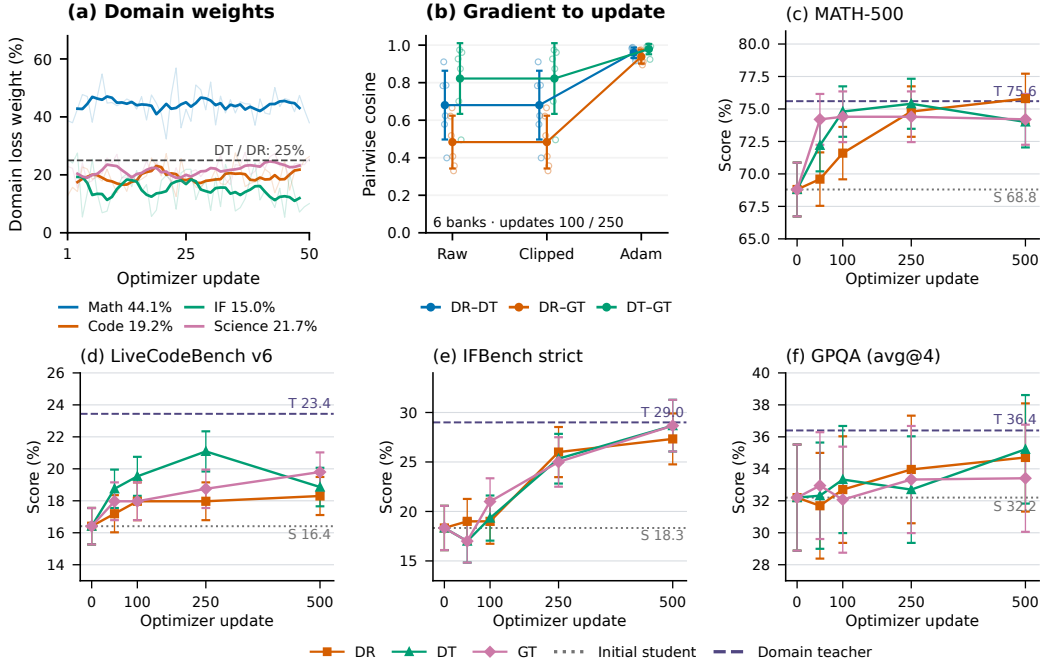


Figure 1: **Loss averaging changes gradient directions and task performance; Adam increases update alignment.** (a) GT domain token shares: individual batches and five-batch averages. (b) Cosines before/after clipping and after Adam (batch means $\pm$ SD). (c–f) I64 task scores (mean $\pm$ std); S/T mark the initial student/teacher.

Table 1: **Endpoint gains coexist with domain-level trade-offs.** Joint-training scores (%); Mean weights domains equally. Δ is the mean difference from PG-DR/Adam (percentage points); brackets give estimated paired 95% intervals[†].

Configuration	MATH-500	Code	IF	GPQA	Mean	Δ [95% interval] [†]
Initial student	68.80	16.41	18.33	32.20	33.94	-4.68 [-6.42, -2.93]
<i>Adam</i>						
PG-DR	73.20	18.27	27.00	35.97	38.61	Reference
PG-DT	73.20	19.24	27.00	34.96	38.60	-0.01 [-1.45, 1.43]
PG-GT	76.20	17.92	27.33	32.94	38.60	-0.01 [-1.44, 1.42]
I16-DR	75.40	17.75	27.00	35.46	38.90	+0.29 [-1.15, 1.73]
I64-DR	75.80	18.30	27.33	34.71	39.04	+0.43 [-1.01, 1.86]
I64-DT	74.00	18.86	28.67	35.22	39.19	+0.58 [-0.87, 2.02]
I64-GT	74.20	19.80	28.67	33.41	39.02	+0.41 [-1.03, 1.85]
<i>SGD</i>						
PG-DR	76.40	19.53	28.33	35.59	39.96	+1.35 [-0.09, 2.79]
PG-DT	75.20	20.31	28.33	33.90	39.44	+0.83 [-0.61, 2.26]
PG-GT	75.00	19.26	25.33	34.71	38.58	-0.04 [-1.47, 1.40]

[†] Estimation procedure and sensitivity checks: Appendix A.3.

Optimizer history also shapes one-step loss changes. Saved-state Adam gives similar held-out KL changes across rules after BF16 writeback (Figure 2c). SGD steps rescaled to the corresponding Adam step norm separate the rules more strongly: all three lower mean KL at update 100 but raise it at update 250.

The average also hides domain differences: at update 250, DT/GT with saved Adam moments lower mean held-out KL but increase instruction-following KL by $0.293/0.272 \times 10^{-3}$.



Figure 2: **DR triggers clipping at every update; Adam moments change step norms and one-step loss reductions.** (a) Retained gradient fraction; labels give clipping rates. (b) FP32 step norms after resetting the first moment or both moments, retaining the step count. (c) Held-out reverse-KL changes (lower is better). Open points are cached batches; filled points and error bars show mean $\pm$ SD.

4 PARAMETER-UPDATE SPARSITY AND TEACHER ALIGNMENT

Concentrated checkpoint changes (Yu et al., 2026) depend on both precision and optimizer history. The joint student changes few BF16 weights but most FP32 master weights; updates from different teachers also align strongly when they share accumulated Adam moments.

4.1 SPARSE BF16 CHANGES MASK WIDESPREAD FP32 UPDATES

For cumulative changes $\Delta = \theta - \theta_0$ or individual steps $u = \theta_{\text{after}} - \theta_{\text{before}}$, we measure the changed fraction $f_\tau(z) = P^{-1} \sum_{j=1}^P \mathbf{1}\{|z_j| > \tau\}$ and the smallest coordinate fraction carrying 90% of squared change. Here P is the number of coordinates; reported nonzero fractions use $\tau = 0$. Both metrics are computed for FP32 master weights and stored BF16 weights.

After 500 updates, joint PG changes 7.30% of BF16 weights, with 3.79% of coordinates carrying 90% of squared change; I64-DR gives 8.46% and 4.38% (Figure 3a). In FP32, however, 97.37% and 97.17% of weights change, and the 90%-energy fractions rise to 34.03% and 35.14%. Even the individual step at update 500 changes 86.80% and 87.05% of FP32 coordinates. Most parameters receive optimizer updates, many of which disappear in BF16. Sparse checkpoint differences therefore do not imply updates restricted to a small parameter subset.

Similar scores also coexist with different cumulative changes: I64-DR/DT/GT change 8.46/10.52/11.24% of BF16 coordinates, with 4.38/5.39/5.92% carrying 90% of squared change. A larger changed fraction does not imply better performance.

4.2 ADAM MOMENTS ALIGN UPDATES FROM DIFFERENT TEACHERS

At the joint-PG update-500 checkpoint, all teacher-specific steps start from identical student parameters and saved Adam moments. We compare teachers on shared prefixes (common inputs) and on their own domains’ prefixes (routed inputs), averaging six teacher pairs within each of four cached batches. With routed inputs and saved Adam moments, mean FP32 update cosines are 0.8774 for PG and 0.8347 for I64. Resetting the first moment while retaining the second, or using norm-matched SGD, reduces both to near zero (Figure 3b). The accumulated first moment accounts for much of the alignment.

4.3 SUBTRACTING THE ZERO-GRADIENT STEP REMOVES MOST UPDATE ALIGNMENT

Adam’s first moment can move parameters even with zero current gradient. For update function $U(g; H)$ and saved state H , we measure the effect of teacher d ’s current gradient by

$$\delta U_d = U(g_d; H) - U(0; H). \quad (6)$$

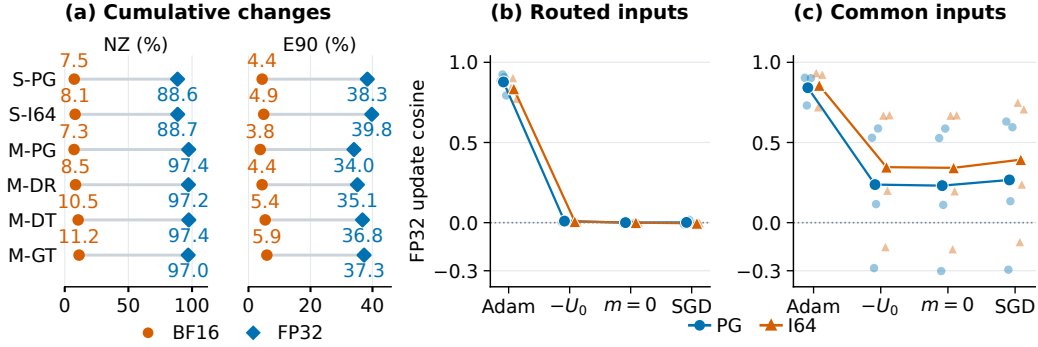


Figure 3: **BF16 parameter changes are sparse; accumulated Adam momentum explains most update alignment on domain-specific inputs.** (a) Nonzero (NZ) and 90%-energy (E90) fractions; S/M: mathematics-only/joint training. M-DR/DT/GT use I64; others use DR. (b,c) FP32 update cosines on domain-specific/shared inputs: $-U_0$ subtracts the zero-gradient Adam step; $m=0$ resets its first moment. Small/large points: batch/overall means.

Both steps start from identical parameters and state; the difference retains Adam’s dependence on its saved moments.

With routed inputs, mean PG/I64 pairwise cosines fall from 0.8774/0.8347 for complete steps to 0.0098/0.0067 for δU_d (Figure 3b). With common inputs, they fall from 0.8415/0.8562 to 0.2373/0.3462 (panel c). Current teacher gradients thus induce step differences with near-zero mean cosine on routed inputs despite strongly aligned complete updates. Their apparent alignment largely reflects shared optimizer history.

5 VOCABULARY SUPERVISION: PARAMETER UPDATES AND TASK PERFORMANCE

More vocabulary supervision need not spread parameter updates more widely. Sampled-token, top- k , and full-vocabulary losses have nearly identical one-step concentration at the same student and Adam state, consistent with concentrated changes under dense distillation (Yu et al., 2026). End-to-end scores nevertheless differ: I64 improves mathematics over PG at the endpoint while losing GPQA accuracy.

5.1 VOCABULARY SUPPORT AND EXPERIMENTAL CONTROLS

PG supervises one sampled token per prefix; I64 uses the intersection of student and teacher top-64 sets (Equations 2, 3). We compare mathematics-only S-PG/S-I64 and joint M-PG/M-I64-DR, sharing initialization, teachers, response averaging, optimizer settings, and prompt schedule within each setting. Joint I16 uses the same objective with $k=16$.

For one-step comparisons, PG, I64, teacher-top-64 (T64), and full-vocabulary reverse KL use identical prefixes, weights, and saved Adam moments at M-PG and M-I64-DR updates 100 and 500. Each checkpoint supplies 96 prefixes in four batches. T64 restricts the PG gradient expectation to the teacher’s fixed top-64 set without renormalizing probabilities.

5.2 UPDATE CONCENTRATION

More supervised tokens leave one-step concentration nearly unchanged. At M-PG/100, all four losses change about 88.0% of FP32 master weights. BF16 changes fall to 0.149% for PG and 0.150% for I64, T64, and full vocabulary. The largest 1% of coordinates carry about 8.8% of FP32 squared step norm and all BF16 squared step norm. BF16 nonzero fractions remain similar across losses at the other checkpoints (Figure 4a). Even zero current gradient changes 0.140% of BF16 parameters at M-PG/100 through Adam’s saved moments.

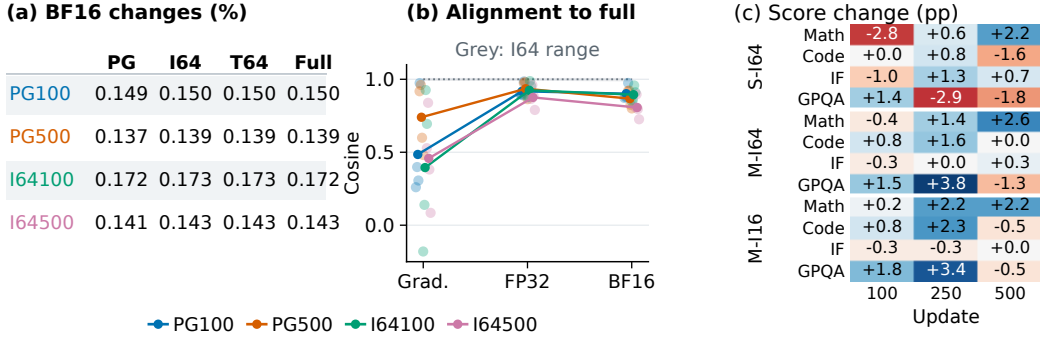


Figure 4: **I64 better approximates the full-vocabulary gradient, with similar BF16 update sparsity and mixed task gains.** (a) Nonzero BF16 changes (%): rows are joint student/update, columns are local losses; colors also identify (b). (b) Cosine to the full-vocabulary gradient or update; shading shows the I64 range. (c) I64/I16 minus PG scores (percentage points), grouped by single-teacher I64, joint I64, and joint I16 training.

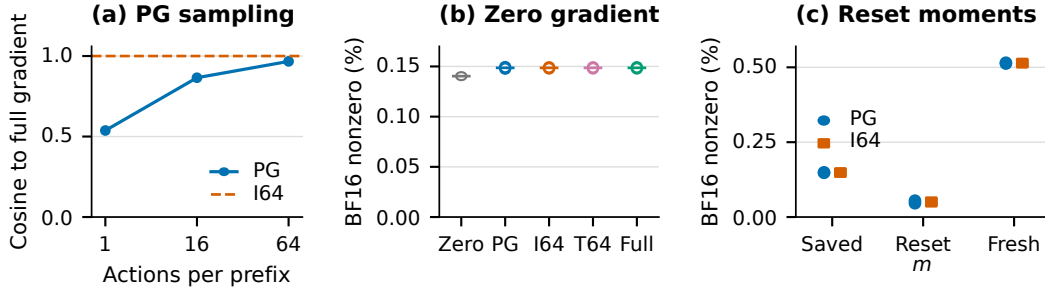


Figure 5: **More token samples increase PG gradient cosine to full-vocabulary KL; Adam momentum alone changes nearly as many BF16 parameters as PG.** (a) Gradient cosine to full-vocabulary KL. (b,c) Nonzero BF16 changes with a zero current gradient or reset moments. Reset- m keeps the second moment and step count; fresh Adam resets both moments and the count. Points denote cached batches; bars show means.

I64 already captures nearly all student mass here: its mean intersection mass exceeds 99.9% at every checkpoint, with a minimum of 96.55% across 384 prefixes. Section 5.4 examines lower-coverage prefixes.

Sampling changes gradient direction while history shapes writeback. Averaging 1, 16, and 64 PG token samples on fixed prefixes raises mean gradient cosine with full-vocabulary KL from 0.537 to 0.865 to 0.967 (Figure 5a). Finite sampling explains part of PG’s gradient difference. The zero-gradient and moment-reset controls in panels b,c show how Adam history changes BF16 writeback.

Single-step similarity also coexists with different cumulative changes: after 500 updates, I64 changes 8.46% of BF16 weights versus PG’s 7.30% (Section 4). Training accumulates changes in the student, generated responses, and optimizer moments that a fixed-state comparison holds constant.

5.3 CAPABILITY UNDER SINGLE-TEACHER AND JOINT TRAINING

I64’s mathematics advantage emerges late and trades off with GPQA. In single-teacher training, PG leads I64 on mathematics by 2.8 points at update 100, but I64 leads by 2.2 at update 500 (Figure 4c). Endpoint I64-minus-PG differences are +0.67 on instruction following, −1.56 on code, and −1.77 on GPQA.

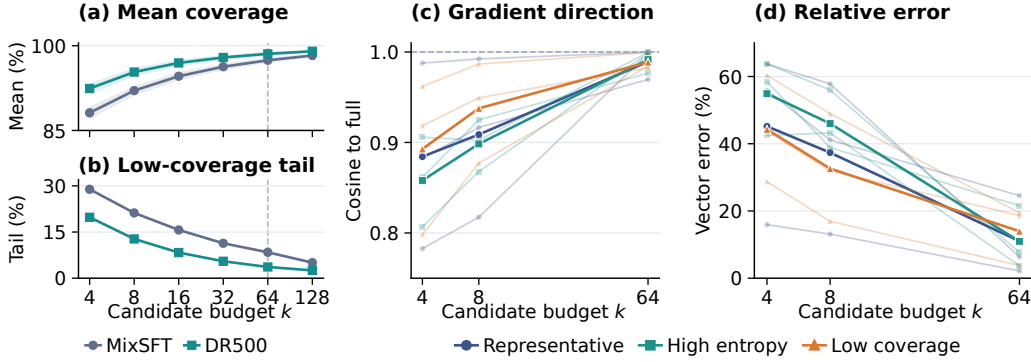


Figure 6: **High coverage and aligned gradients can conceal omitted mass and vector error.** (a,b) Responses sampled separately from MixSFT and DR500, 384 each; band: response-cluster 95% interval; tail: mass below 90%. Dashed lines mark $k = 64$. (c,d) Fixed DR500 gradients: three batches per stratum (thin), with means (bold). Coverage and teacher targets use BF16 forwards; student gradients use FP32.

Joint-training endpoint differences are +2.60 on mathematics, +0.03 on code, +0.33 on instruction following, and -1.26 on GPQA (Table 1). The mean difference is +0.43 points, with an estimated paired 95% interval of $[-1.01, 1.86]$ (Table 5). Rankings also change: at update 250, I64 leads on mathematics, code, and GPQA and ties on instruction following. Its mean advantage then shrinks from 1.69 to 0.43 points by update 500.

Top-64 does not consistently outperform top-16. Joint I16 and I64 endpoint means are 38.90% and 39.04%, respectively. I64 gains 0.13 mean points on these benchmarks. On separate test sets of 256 questions per domain, their means are also close, at 44.56% and 44.46%: I64 gains 1.96 points on mathematics but loses 2.35 on code. Increasing k changes both candidate support and token weights in Equation 3, producing task-dependent trade-offs.

5.4 COVERAGE, GRADIENT ERROR, AND CAPABILITY IN SMOLLM3

SmolLM3-3B provides a lower-coverage comparison using the MixSFT initialization, its I64-DR update-500 student, and three Open-MOPD teachers for mathematics, code, and instruction following (Gao et al., 2026).

High mean coverage leaves a measurable tail. At $k = 64$, mean intersection mass is 97.42% for MixSFT and 98.56% for DR500 on their respective generated responses. Yet 8.42% and 3.65% of domain/response-balanced prefix weight falls below 90% mass (Figure 6a,b). Increasing k reduces this tail without eliminating it.

Direction alone understates approximation error. On representative, high-entropy, and low-I64-coverage strata at fixed DR500 parameters, I64/full-KL gradient cosines range from 0.9696 to 0.9998 across nine batches. Nevertheless, relative vector error $\|g_{I64} - g_{full}\|/\|g_{full}\|$ spans 2.16–24.54% (Figure 6c,d). I4 and I8 have larger errors than I64, but the low-coverage stratum is not uniformly worst. High average mass and cosine can therefore hide substantial gradient approximation error.

Equal endpoint scores can hide different solved questions. We compare MixSFT with update-500 I64-DT/Adam and PG-DR/SGD using shared prompts, scorers, and decoding settings within each benchmark. MATH-500 greedy accuracy is 50.60% for MixSFT and 53.80% for PG-DR/SGD. AIME pass@8 is the fraction of questions with at least one correct answer among eight samples. In 2024, the three models solve 3/30, 8/30, and 5/30 questions, respectively; in 2025, each solves 4/30. Yet their 2025 average response accuracies are 1.67/3.75/2.08%, and only one solved question is common to all three.

Table 2: **The retention constraint leaves every checkpoint choice unchanged.** Test Δ is the selected-checkpoint mean minus the update-500 mean (percentage points).

Trajectory	Endpoint	Max dev. mean	Retention	Test Δ (pp)
Other nine trajectories	500	500	500	0.000
PG-DT / Adam	500	250	250	-0.366

6 PRACTICAL IMPLICATIONS AND CHECKPOINT SELECTION

Loss weights and optimizer state require separate measurements. Figure 1 shows that equal prompt counts yield unequal token loss weights, while DT and DR still produce different gradients. Figure 3 separates FP32 changes from BF16 sparsity and current teacher signals from Adam history. Figures 4 and 6 show why coverage and update similarity should be read alongside per-task scores and gradient error.

A retention constraint leaves checkpoint selection unchanged. We compare update 500 with two development-based rules over updates 50/100/250/500 for Adam and 100/250/500 for SGD: maximize the equal-domain mean, or maximize it subject to per-domain retention,

$$s_d(\theta) \geq s_d(\theta_0) - \epsilon_d \quad \text{for all } d, \quad \epsilon_d = 2 \text{ percentage points.} \quad (7)$$

Here s_d is the development score for domain d and θ_0 the initial student. Both rules keep the initial student unless the selected mean improves by at least one point. Choices are fixed before evaluation on the separate test sets of 256 questions per domain.

Table 2 summarizes the selection experiment. Both development rules choose update 500 in nine trajectories and update 250 for PG-DT/Adam. The latter scores 43.14% on the additional test, below the final checkpoint’s 43.51%; the retention constraint and earlier selection provide no observed test-score benefit. Rankings also change across evaluation sets: PG-DR/SGD exceeds Adam by 1.35 mean points on the development benchmarks but is 0.07 points lower on the additional test.

7 RELATED WORK

Loss normalization. DAPO replaces response-level with token-level averaging (Yu et al., 2025); Dr. GRPO analyzes response-length and reward-normalization bias (Liu et al., 2025). We distinguish domain weighting from within-domain response weighting and measure their effects on gradients, Adam updates, and task scores.

Multi-teacher distillation. MOPD integrates specialized teachers on student-generated responses (Ma et al., 2026). Open-MOPD introduces domain token balancing, performance-gap weighting, and reward refresh (Gao et al., 2026). Its token-share correction corresponds to DT for fixed batches and domain weights; our comparison further isolates response-length weighting.

Post-training parameter geometry. Mukherjee et al. (2025) study small parameter subsets supporting RL fine-tuning. Yu et al. (2026) report concentrated changes under dense distillation and overlap between separately trained students. We compare teachers at the same jointly trained student and Adam state, separating FP32 from BF16 changes. Shen et al. (2026) study OPD geometry and response-position selection; our supervision comparison varies vocabulary candidates at each position.

8 DISCUSSION

Loss construction, optimizer history, and numerical precision each shape how MOPD updates appear. Adam’s accumulated moments align otherwise distinct teacher signals, while BF16 storage hides dense FP32 changes. Loss normalization changes gradients more than final top-64/Adam scores, and increasing vocabulary supervision produces task- and checkpoint-dependent gains. SmolLM3 further shows that high coverage and cosine can coexist with substantial tail mass and

vector error. These results support evaluating distillation objectives through per-task learning curves together with current gradients, optimizer updates, and stored-weight changes.

REPRODUCIBILITY STATEMENT

The evidence bundle includes figure and table values, configurations, source hashes, scoring audits, and plotting scripts. The appendix provides full protocols, supporting analyses, and overlap checks.

ETHICS STATEMENT

We use public models and benchmarks. Generative AI assisted manuscript editing and analysis code; the authors are responsible for the measurements and their interpretation.

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Table 3: **PG/I64 performance depends on the task and checkpoint.** Mean scores (%): S is mathematics-only training and M is joint training; both use DR. Joint endpoints appear in Table 1.

Configuration	Updates	Math	Code	IF	GPQA
S-PG	100	77.40	18.75	18.67	31.82
S-PG	250	73.80	20.31	17.00	35.35
S-PG	500	73.80	19.53	18.00	31.82
S-I64	100	74.60	18.75	17.67	33.21
S-I64	250	74.40	21.09	18.33	32.45
S-I64	500	76.00	17.97	18.67	30.05
M-PG	50	69.60	16.41	18.67	28.03
M-PG	100	72.00	17.19	19.33	31.19
M-PG	250	73.40	16.41	26.00	30.18
M-I64-DR	50	69.60	17.19	19.00	31.69
M-I64-DR	100	71.60	17.97	19.00	32.70
M-I64-DR	250	74.80	17.97	26.00	33.96

A EXPERIMENTAL PROTOCOL AND RESULTS

A.1 MODEL, DATA, AND TRAINING PROTOCOL

The main-study student is Qwen3-1.7B (Yang et al., 2025); training uses seeds 42, 43, and 44. The four teachers were trained by reinforcement learning from the same initial model, specializing in mathematics, code, instruction following, and science, respectively. Student and teachers share the tokenizer and native chat template with thinking disabled. The experiment records identify the model revisions and weight files.

AdamW configurations use learning rate 2.5×10^{-7} with $(\beta_1, \beta_2) = (0.9, 0.98)$, $\epsilon = 10^{-8}$, no weight decay or schedule, and global norm-1 clipping. Training sampling uses temperature 1 and top- $p = 1$, with a 2,048-token prompt cap and 4,096-token response cap. One fresh response is sampled per prompt. Each optimizer update aggregates 64 responses; domain allocation is specified below.

The multi-teacher setup allocate 16 prompts to each domain per update and use target domain weights $1/4$. Single-teacher setup allocate all 64 prompts to the same domain. I64 uses the top-64 objective in Equation 3; I16 uses $k = 16$.

A.2 EVALUATION AND SCORING

Each main-study capability suite contains MATH-500 (500 questions) (Hendrycks et al., 2021), LiveCodeBench v6 (1,055 questions) (Jain et al., 2024), IFBench strict (300 questions) (Pyatkin et al., 2025), and GPQA Diamond (198 questions) (Rein et al., 2023). Mathematics, code, and instruction following use one greedy response per question. GPQA uses four responses per question at temperature 1 and top- $p = 1$. Its score is the mean correctness over all four responses, then over questions: average@4 accuracy. Evaluation permits up to 32,768 response tokens, bounded by the remaining configured 32,768-token context. Each complete suite contains 2,053 questions and 2,647 responses per model evaluation.

The expanded development grid comprises updates of 50/100/250/500 for seven Adam trajectories and 100/250/500 for three SGD trajectories, plus the initial student evaluation. Appendix G.1 describes additional test sets for the same four domains. Table 3 supplements the joint endpoints in Table 1 with single-teacher scores and intermediate PG/I64 checkpoints.

The four RL teachers were evaluated separately on their respective domain benchmarks using the same prompt and decoding protocol (Table 4). These scores quantify each teacher’s performance on its target domain.

Table 4: **Reference scores for the four domain-specific teachers (%)**. Science uses GPQA average@4.

Teacher / target domain	Responses	Score (%)	Truncation (%)
Math	500	75.60	6.80
Code	1055	23.44	24.22
If	300	29.00	7.33
Science	792	36.40	0.51

A.3 ESTIMATED PAIRED UNCERTAINTY FOR THE QWEN ENDPOINTS

Status and scope. The intervals in Table 1 and Table 5 are **model-based estimates**, obtained by a Gaussian parametric bootstrap calibrated to archived paired-question intervals. They are not nonparametric bootstrap intervals computed from the current Table 1 question outcomes: those outcomes are unavailable in this manuscript workspace. We retain the reported domain scores and estimate the variance of their paired differences. No question-level outcomes or additional training runs are reconstructed. This procedure differs from the empirical question bootstrap used for the SmolLM3 comparisons in Appendix H.5.

Calibration from existing paired comparisons. The archived `paired_comparisons.json` and `capability.json` (revision 74219fe52df5) contain earlier paired 95% intervals, obtained using 10,000 question-bootstrap replicates with seed 1042. Their GPQA bootstrap retains all four responses to each question. We use the latest available comparison for each joint-training pair: I64-DR versus I64-DT/GT at update 500 and PG versus I64-DR at update 250. Comparisons with initialization use each of the three I64 endpoints and PG at update 250. This gives three method-comparison and four initialization-comparison calibration records per domain. Shared questions make these records dependent; we use their median as a working calibration, not as independent observations for an additional significance test.

For archived record r in domain d , let $p_{A,r}, p_{B,r}$ denote score proportions, n_r the question count, and $[L_r, U_r]$ its interval in percentage points. We approximate its standard error by its interval width and define

$$\kappa_{d,c} = \text{median}_{r \in \mathcal{R}_{d,c}} \frac{n_r [(U_r - L_r) / (200 z_{0.975})]^2}{p_{A,r}(1 - p_{A,r}) + p_{B,r}(1 - p_{B,r})}, \quad z_{0.975} = 1.959964, \quad (8)$$

where c distinguishes method comparisons from comparisons with initialization. The resulting variance ratios are

Calibration	Math	Code	IF	GPQA
Method versus method	0.355	0.062	0.105	0.189
Method versus initialization	0.374	0.147	0.414	0.197

These are effective variance ratios, not measured correlations for the current endpoints. For GPQA they incorporate both the four-response question average and cross-model dependence. The archived code evaluation used 128 questions; transferring its ratio to the current 1,055-question benchmark is an assumption. Applying the same method-comparison calibration to SGD and I16 also assumes comparable paired variability. The calibration does not identify uncertainty in these transfers.

Parametric bootstrap of the paired differences. For current score proportions $p_{A,d}, p_{B,d}$, set $\delta_d = p_{A,d} - p_{B,d}$ and

$$V_{0,d} = \frac{p_{A,d}(1 - p_{A,d}) + p_{B,d}(1 - p_{B,d})}{n_d}, \quad (n_d) = (500, 1055, 300, 198). \quad (9)$$

We scale this independent Bernoulli reference by $\kappa_{d,c}$ and enforce feasible bounds for question-level differences. In particular, binary correctness requires $V_d \geq (|\delta_d| - \delta_d^2)/n_d$, preventing an assumed correlation from producing an impossibly narrow interval. More generally, with score

Table 5: **Small endpoint contrasts remain uncertain under the calibrated model.** Mean differences (percentage points) and estimated paired 95% intervals; 2κ weakens the assumed variance reduction, and $\kappa = 1$ uses the independent Bernoulli reference. These are pointwise, model-based intervals.

Contrast	Δ	Calibrated	$\min(2\kappa, 1)$	$\kappa = 1$
I64/Adam: DT – DR	+0.15	[-1.28, 1.59]	[-1.88, 2.18]	[-3.20, 3.51]
I64/Adam: GT – DR	-0.02	[-1.44, 1.41]	[-2.04, 2.01]	[-3.36, 3.33]
I64/Adam: DT – GT	+0.17	[-1.27, 1.61]	[-1.87, 2.20]	[-3.19, 3.53]
PG/Adam: DT – DR	-0.01	[-1.45, 1.43]	[-2.05, 2.03]	[-3.37, 3.35]
PG/Adam: GT – DR	-0.01	[-1.44, 1.42]	[-2.04, 2.01]	[-3.35, 3.33]
PG/Adam: DT – GT	+0.00	[-1.43, 1.43]	[-2.02, 2.02]	[-3.33, 3.34]
PG/SGD: DT – DR	-0.53	[-1.96, 0.90]	[-2.55, 1.49]	[-3.88, 2.83]
PG/SGD: GT – DR	-1.39	[-2.82, 0.04]	[-3.41, 0.63]	[-4.73, 1.96]
PG/SGD: DT – GT	+0.86	[-0.57, 2.29]	[-1.16, 2.88]	[-2.48, 4.20]
PG-DR: SGD – Adam	+1.35	[-0.09, 2.79]	[-0.68, 3.39]	[-2.01, 4.71]
PG-DT: SGD – Adam	+0.84	[-0.60, 2.27]	[-1.20, 2.87]	[-2.52, 4.19]
PG-GT: SGD – Adam	-0.02	[-1.44, 1.40]	[-2.03, 1.99]	[-3.34, 3.30]
Adam/DR: I64 – PG	+0.43	[-1.01, 1.86]	[-1.61, 2.46]	[-2.93, 3.78]
Adam/DR: I64 – I16	+0.13	[-1.30, 1.56]	[-1.89, 2.15]	[-3.21, 3.47]

spacing $h_d = 1$ for greedy tasks and $h_d = 1/4$ for GPQA, let $a_d = h_d \lfloor \delta_d / h_d \rfloor$. We use

$$V_{\min,d} = \frac{(\delta_d - a_d)(a_d + h_d - \delta_d)}{n_d}, \quad (10)$$

$$V_{\max,d} = \frac{p_{A,d} + p_{B,d} - 2 \max(0, p_{A,d} + p_{B,d} - 1) - \delta_d^2}{n_d}, \quad (11)$$

$$\hat{V}_d = \min\{V_{\max,d}, \max(V_{\min,d}, \kappa_{d,c} V_{0,d})\}. \quad (12)$$

The GPQA unit remains a question with its four-response average; we do not use 792 independent responses. We generate $\delta_d^{*(b)} = \delta_d + \sqrt{\hat{V}_d} Z_d^{(b)}$, with independent standard-normal draws across domains. There are 100,000 replicates (seed 1042; 50,000 draws and their negatives to preserve symmetry). Each replicate’s four-domain contrast is $25 \sum_d \delta_d^{*(b)}$ percentage points. Its 2.5th and 97.5th percentiles give the reported interval. Thus the simulation is a Gaussian approximation to the paired sampling distribution, conditional on the transferred variance ratios; it does not resample observed current questions. All contrasts use the displayed domain scores before rounding, and a model compared with itself has exactly zero difference and variance.

Sensitivity and interpretation. Table 5 doubles each calibrated ratio, capped at one, and also sets every ratio to one. Under the calibrated model, the mean-contrast interval widths are about 43% of the independent reference for method comparisons. Nevertheless, all listed method-comparison intervals include zero: narrower intervals do not establish either a method advantage or equivalence. All ten improvements over initialization remain positive even under the independent reference; their point estimates range from 4.64 to 6.03 points, with the smallest lower bound 1.37 points. These are pointwise, unadjusted intervals conditional on the evaluated configurations, without training-seed variance, calibration uncertainty, or a correction for checkpoint selection. They do not establish a population ranking across training runs.

The marginal error bars retained for absolute Qwen learning curves in Figures 1 and 9 use the aggregate-score approximation $100\sqrt{p_d(1-p_d)/n_d}$, propagated across equally weighted domains for the mean. They are estimated standard deviations, not measured training-seed variation, and are not used to assess paired contrasts. Diagnostic batch error bars retain their separately stated definitions.

B LENGTH-WEIGHTING DERIVATION

Fix a rollout batch, its valid-token masks, and domain loss weights λ_d . The PG log-ratio coefficients are treated as constants during backpropagation. For $n_d = |\mathcal{B}_d|$, let $\bar{T}_d = n_d^{-1} \sum_{i \in \mathcal{B}_d} T_i$ and

Table 6: **Equal domain weights do not imply equal gradient directions.** Raw-gradient cosines on identical cached responses. DR and DT use equal domain loss weights; only DT weights responses by length.

Updates	Batch	DR-DT	DR-GT	DT-GT
100	1	0.7838	0.6219	0.9266
100	2	0.9110	0.5180	0.7010
100	3	0.5780	0.6659	0.9745
250	1	0.6240	0.4072	0.8742
250	2	0.7864	0.3302	0.4953
250	3	0.3990	0.3569	0.9612

$\bar{g}_d = n_d^{-1} \sum_{i \in \mathcal{B}_d} g_i$. Define the empirical covariance using the same $1/n_d$ normalization:

$$\text{Cov}_d(T, g) = \frac{1}{n_d} \sum_{i \in \mathcal{B}_d} (T_i - \bar{T}_d)(g_i - \bar{g}_d). \quad (13)$$

Expanding gives $n_d^{-1} \sum_i T_i g_i - \bar{T}_d \bar{g}_d$. Dividing by $\bar{T}_d$ gives Equation 5. GT combines the resulting domain gradients using observed token shares; DT combines them using λ_d ; DR combines $\bar{g}_d$ using λ_d . Multiplying each domain’s globally averaged terms by its target weight divided by its observed token share therefore gives DT exactly.

The derivation is a finite-batch weighted-average identity. Response sampling, masks, and lengths are held fixed. Its population analogue involves an expectation of a ratio; replacing that expectation by a ratio of expectations requires additional assumptions or a large-batch limit.

B.1 FIXED-BATCH GRADIENT COMPARISON

At M-PG checkpoints 100 and 250, three cached response batches (seeds 1042, 1043, and 1044) each contain eight responses per domain. Responses use the training prompt format and 4,096-token response cap. For each batch, we compute each response’s mean I64 gradient once at the saved BF16 model values using FP32 computation, then aggregate those gradients under DR, DT, and GT. All three averaging rules use the same responses, token masks, teacher scores, and model coordinates. Each comparison starts from the same student parameters and Adam first and second moments.

Table 6 reports every pairwise raw-gradient cosine. The domain-wise covariance decomposition is checked in FP64 for all 24 batch-domain groups; the largest relative residual is 1.53×10^{-15} . This numerical check verifies the aggregation identity. The measured cosine differences show that response length and gradient direction are correlated in these batches.

The optimizer interventions in Figure 2 reuse these six batches. Each aggregate gradient is globally clipped at norm 1 and applied to the same saved FP32 Adam state for that batch. The saved-state branch retains both moments; the reset- m branch zeros the first moment while retaining the second moment and optimizer step counter; reset- (m, v) zeros both moments while retaining the step counter. The momentum-free SGD control follows the negative gradient and is rescaled separately for DR, DT, and GT to match the FP32 Adam step norm for that averaging rule. After each FP32 update, the model parameters are cast to BF16 and evaluated on one held-out response per domain. Evaluation uses full-vocabulary reverse KL; the experiment archive also retains the I64 surrogate used for training. Each comparison applies a single optimizer step to a checkpoint copy.

C PARAMETER AND OPTIMIZER MEASUREMENTS

C.1 COORDINATES AND RECORDED QUANTITIES

Training measurements separately record gradients before and after clipping, FP32 optimizer steps, and cumulative parameter changes in the FP32 master weights and BF16 model weights. Cumulative changes are measured from the same initialization. All global statistics use unique model coordinates, excluding vocabulary padding and duplicate entries for tied weights. Per-layer statistics and

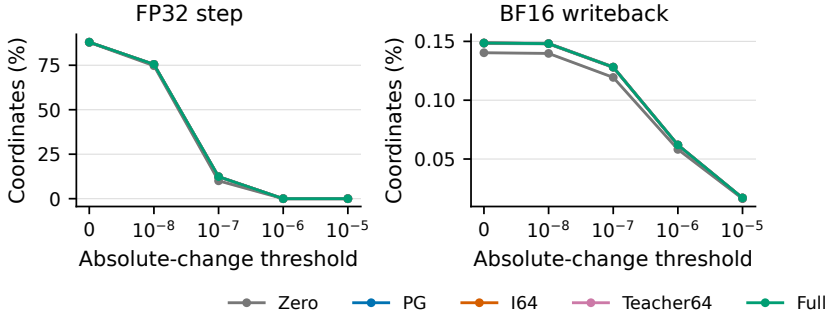


Figure 7: **Update sparsity depends on the numerical threshold.** FP32 steps and BF16 parameter changes; lines show batch means and shading their range.

absolute-threshold sensitivities are retained in the experiment archive. For an absolute threshold τ , sparsity is the fraction of coordinates with magnitude at most τ ; the figures show the fraction exceeding τ . Energy concentration uses squared Euclidean norm and ranks coordinates by absolute change.

For checkpoint-based measurements, the exported checkpoint contains FP32 master values, BF16 model values, first and second moments, optimizer hyperparameters, and step counters. Student and teacher values are verified against their saved models in the same precision. Forward and backward passes evaluate these values in FP32 with TF32 disabled. All proposed steps are computed from the same saved state with a non-fused Adam implementation. The online training optimizer uses its own fused implementation.

We compare three Adam states at the same checkpoint. *Saved* retains both moments and the step counter. *Reset m* zeros the first moment while retaining the second moment and step counter. *Fresh* zeros both moments and resets the step counter. Global norm clipping at 1 is applied before the Adam calculation, and weight decay is zero. The plotted controls use two cached response batches; the zero-gradient control is evaluated once. For the zero-gradient branch, the saved first moment continues to decay and move the master weights. Casting the saved FP32 master weights to BF16 without an optimizer step exactly reproduces the saved model. Figure 7 shows how the threshold changes the measured fraction of updated coordinates.

Setting the current gradient to zero while retaining the trained Adam moments changes 0.1403% of BF16 coordinates, compared with 0.1486% for PG. Resetting the first moment gives 0.0504% for PG; resetting both moments and the step counter gives 0.5140%. The parameters changed by Adam therefore depend on its accumulated moments and BF16 rounding, not just on which entries of the current gradient are nonzero. These controls supplement the update-sparsity measurements in Sections 4 and 5.

Figure 5(b,c) displays these BF16 controls. The full FP32 and BF16 measurements are retained in the experiment archive.

C.2 CACHED RESPONSES, GRADIENT OVERLAP, AND GRADIENT NORMS

These raw-gradient diagnostics use the M-PG/100 student, whereas Figure 3(b,c) measures teacher-conditioned optimizer steps at M-PG/500. Batches use seeds 42 and 43, one response per domain per batch, a 2,048-token prompt cap, and a 256-token generation cap. All eight responses reach the cap. Four response positions are sampled without replacement per response. Each response’s four prefixes receive equal weight. For domain-specific inputs, each teacher scores the response from its own domain; for shared inputs, each teacher scores all four responses with equal domain weights. A third control evaluates every teacher on every input domain separately, keeping four prefixes per gradient. These conditions reuse the same responses, allowing paired comparisons within each batch.

Let $S_d(\rho)$ contain the largest-magnitude ρ fraction of teacher d ’s gradient or update coordinates, with stable coordinate-index tie breaking. Jaccard overlap $J(d, e) = |S_d(\rho) \cap S_e(\rho)| / |S_d(\rho) \cup S_e(\rho)|$

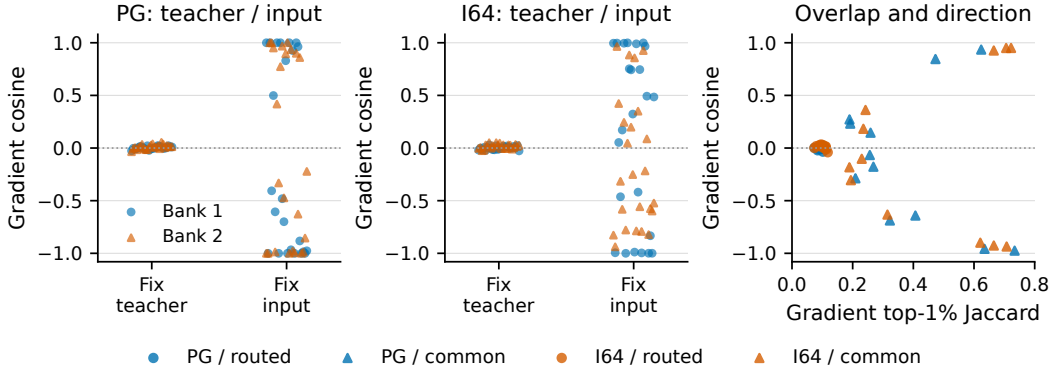


Figure 8: **Gradient overlap depends strongly on the input domain.** Left/center: teacher-by-input-domain controls. Right: top-1% gradient Jaccard versus cosine on domain-specific or shared inputs.

Table 7: **Vocabulary supervision produces similar single-step update concentration.** M-PG/100 means over four cached batches. NZ: nonzero coordinates (%); top-1%: squared step norm in the largest 1% of coordinates (%).

Loss	FP32 NZ	BF16 NZ	FP32 top-1%	BF16 top-1%
PG	87.982	0.149	8.845	100.000
I64	87.989	0.150	8.827	100.000
T64	87.989	0.150	8.828	100.000
Full	87.999	0.150	8.828	100.000

measures shared locations; cosine also captures signs and magnitudes (Yu et al., 2026). Table 8 reports PG gradient norms for domain-specific inputs; the first batch’s instruction-following gradient has norm 6.13×10^{-9} . Since cosine divides out gradient norms, a negative value need not imply a large opposing contribution. Independent uniform sets of fraction ρ have expected-intersection to expected-union ratio $\rho/(2 - \rho)$, approximately 0.00503 at $\rho = 1\%$. This reference uses uniform coordinates; layer and weight-magnitude structure remain in the measured gradients.

The largest-magnitude 1% of raw-gradient coordinates give mean Jaccard overlaps of 0.0951 for PG and 0.0999 for I64 on domain-specific inputs, averaging the two batches and six teacher pairs. Their ranges are 0.0744–0.1077 and 0.0737–0.1178. When every teacher instead scores all four domain responses, the respective means are 0.3808 and 0.4577. The latter comparison changes both input composition and prefix count; the teacher-by-input-domain comparison below fixes the prefix count.

Holding the teacher fixed while changing the input domain gives mean gradient cosines of 0.0059 for PG and 0.0072 for I64. Holding the input domain fixed while changing teachers produces both aligned and opposed gradients. For domain-specific inputs the mean cosines are -0.0020 and 0.0081 , respectively, despite partial coordinate overlap. Figure 8 shows how input choice affects the teacher-overlap measurements in Section 4.

D VOCABULARY SUPERVISION AND COVERAGE

Single-step comparison using four response batches. The M-PG/100 row of Figure 4(a) uses four cached response batches with seeds 2042–2045. Each batch contains one response per domain. Generation uses the training prompt format, a 2,048-token prompt cap, and a 4,096-token response cap. The 16 response lengths range from 81 to 4,096 tokens; two responses reach the cap. Six positions per response give 96 prefixes. PG actions are independently sampled from the student at these fixed prefixes. All losses use the same prefixes, equal domain weights, and equal response weights within each domain. Gradients are globally clipped at norm 1 before applying the same saved Adam state. Table 7 adds the FP32 and energy-concentration measurements for this state.

Table 8: **The teacher top-64 set captures nearly all student probability mass in these controls.** Means over sampled prefixes; gradient norms use domain-specific inputs. Entropy is in nats, and 100% may reflect rounding.

Bank	Domain	Mass in T (%)	Mass in I (%)	Entropy	$\ g_{PG}\ _2$
1	Math	100.0000	100.0000	0.3243	132.6
1	Code	99.9968	99.9965	0.7701	163.8
1	IF	100.0000	100.0000	1.85×10^{-5}	6.13×10^{-9}
1	Science	99.9184	99.9162	0.7405	54.82
2	Math	99.9991	99.9991	0.3221	2.605
2	Code	100.0000	100.0000	0.2962	3.167
2	IF	100.0000	100.0000	0.5004	55.52
2	Science	99.9993	99.9992	0.2442	0.8505

The loss and Adam implementations match the two-batch controls in Appendix C. Appendix G.3 reports the actual I64 intersection coverage at all four student checkpoints.

Implementation details. For I64 (Equation 3), candidate sets and teacher scores are fixed at roll-out. Each student forward recomputes current probabilities at the saved candidate positions and refreshes the detached coefficients. PG also refreshes its detached log-ratio coefficients and uses no advantage or policy-ratio clipping.

Teacher-top-64 reference objective. The teacher-top-64 diagnostic uses the fixed teacher set T and full-vocabulary probabilities:

$$\ell_{T64} = \sum_{v \in T} \left[p_{\theta}(v) \log \frac{p_{\theta}(v)}{q_d(v)} - p_{\theta}(v) + q_d(v) \right]. \quad (14)$$

Its gradient is $\sum_{v \in T} p_{\theta}(v) \log(p_{\theta}(v)/q_d(v)) \nabla_{\theta} \log p_{\theta}(v)$. I64 instead uses the detached weights in Equation 3, normalized over the original student set S . T64 and full-vocabulary KL are used only for single-step comparisons. End-to-end training compares PG, I16, and I64; the I16 runs have no corresponding single-step gradient or update measurements.

Differentiating Equation 2 gives

$$\nabla_{\theta} \ell_{\text{full}} = \sum_{v \in \mathcal{V}} p_{\theta}(v) \left(\log \frac{p_{\theta}(v)}{q_d(v)} + 1 \right) \nabla_{\theta} \log p_{\theta}(v) = \mathbb{E}_{a \sim p_{\theta}} \left[\log \frac{p_{\theta}(a)}{q_d(a)} \nabla_{\theta} \log p_{\theta}(a) \right] \quad (15)$$

using $\sum_v \nabla_{\theta} p_{\theta}(v) = 0$. This identity conditions on fixed prefixes and weights; the training loss additionally depends on the sampled response and the token weights induced by its final length and the normalization rule. Table 8 reports candidate-set coverage for the two-batch gradient-overlap and Adam-state controls.

E SUPPORTING MEASUREMENTS FOR THE MAIN FIGURES

Teacher-update controls. Figure 3(a) compares cumulative parameter changes at update 500 under DR, with additional I64-DT/GT rows; paired points distinguish FP32 and BF16 for each statistic. Panels (b,c) start from the same M-PG/500 student parameters and Adam moments. With domain-specific inputs, each teacher scores responses from its own domain; with shared inputs, all teachers score the same prefixes. Each of four cached response batches contains six teacher pairs. Small points summarize pairs within a batch, and large points average batches. Equation 6 subtracts the Adam step with the current gradient set to zero before computing FP32 update cosines. It subtracts the zero-gradient baseline; the remaining difference still depends nonlinearly on the saved moments. SGD and reset- m steps are individually rescaled to the norm of the Adam step with trained moments; reset- m retains the second moment and step counter. Teacher pairs reuse teachers and responses, so pairwise observations are dependent. These calculations use non-fused PyTorch updates on copies of saved checkpoints and need not reproduce the fused training optimizer bit for bit. Mathematics-only and joint training also differ in mathematics prompt exposure (Appendix A).

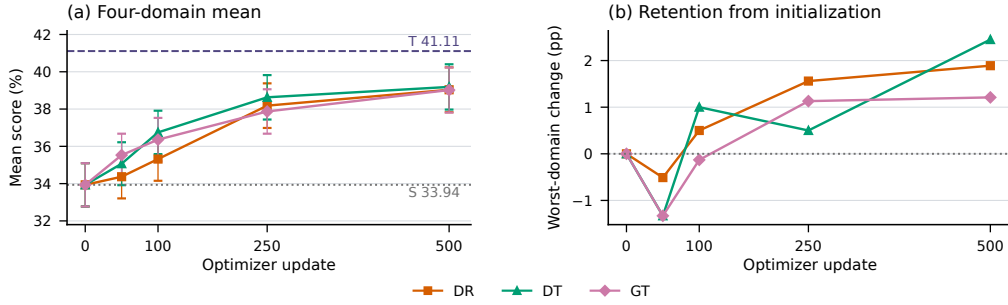


Figure 9: **Mean performance gains can coexist with a regression in an individual domain.** (a) Four-domain mean $\pm$ std; S/T mark the initial student/teacher. (b) Worst-domain change from initialization; zero means no regression. Lines connect reported point estimates.

Vocabulary-supervision controls. Figure 4(a,b) uses four cached response batches at each of four student checkpoints: M-PG and M-I64-DR at updates 100 and 500. Losses use identical prefixes and Adam moments within each checkpoint, while responses can differ between checkpoints. Panel (a) reports batch means. In panel (b), points are individual batches, lines connect PG batch means, and shading spans the I64 measurements. Direct pairwise comparisons of T64 update directions were not recorded; I16 appears only in the end-to-end training results. Panel (c) compares S-I64, M-I64, and M-I16 with PG at the same task, teacher setting, and update. Table 7 adds FP32 and energy-concentration measurements for the M-PG/100 comparison in Appendix D.

E.1 ADDITIONAL OPTIMIZER AND SAMPLING CONTROLS

Figure 2 uses the six cached response batches and training clipping records described in Appendix B.1. In panel (a), boxes mark the interquartile range and whiskers the 5th–95th percentiles; labels give the fraction of training updates clipped. *Reset- m* and *reset- (m, v)* retain the optimizer step counter. Panel (c) averages four domain probe losses within each batch; error bars show sample standard deviations over the three batches at each checkpoint.

Figure 5(a) uses three M-PG/100 response batches, with a 4,096-token response cap, containing 48 responses and 288 prefixes. Twelve fixed subsets of prefixes each receive two independent draws of 1, 16, or 64 PG actions per prefix. I64 is evaluated once per subset. Increasing the sample count raises the mean PG gradient cosine to full-vocabulary KL from 0.5374 to 0.8652 and 0.9667; I64 reaches 0.999988. These repeats measure sampling variation at a fixed checkpoint. Panels (b,c) use the separate two-batch experiment with a 256-token response cap (Appendix C); its zero-gradient control is evaluated once. *Reset- m* retains the second moment and step counter, whereas fresh Adam resets both moments and the counter. The sampling and moment-reset experiments use different response batches and are not paired.

Cumulative parameter changes. Figure 10 shows the trajectories underlying the update-500 summary in Figure 3(a).

F PROMPT EXPOSURE AND CANDIDATE-SET WEIGHTS

When each domain loss is averaged separately, its prompt quota and loss weight control different quantities. Increasing the quota at fixed λ_d changes the number of sampled responses and the variance of the gradient estimate, without directly increasing the domain’s expected contribution to the total gradient. Mathematics-only runs receive four times as many mathematics prompts per update as joint runs. Their score differences therefore cannot be attributed to teacher count alone.

Candidate-set coverage also scales the distillation gradient. For the student candidate set S and its intersection I with the teacher set in Equation 3, the retained student weights sum to $r(h) = p_\theta(I | h)/p_\theta(S | h)$. At a fixed prefix with a nonempty intersection and detached coefficients, the I64 gradient equals $r(h)$ times the gradient obtained by renormalizing the same weights over I . Equal

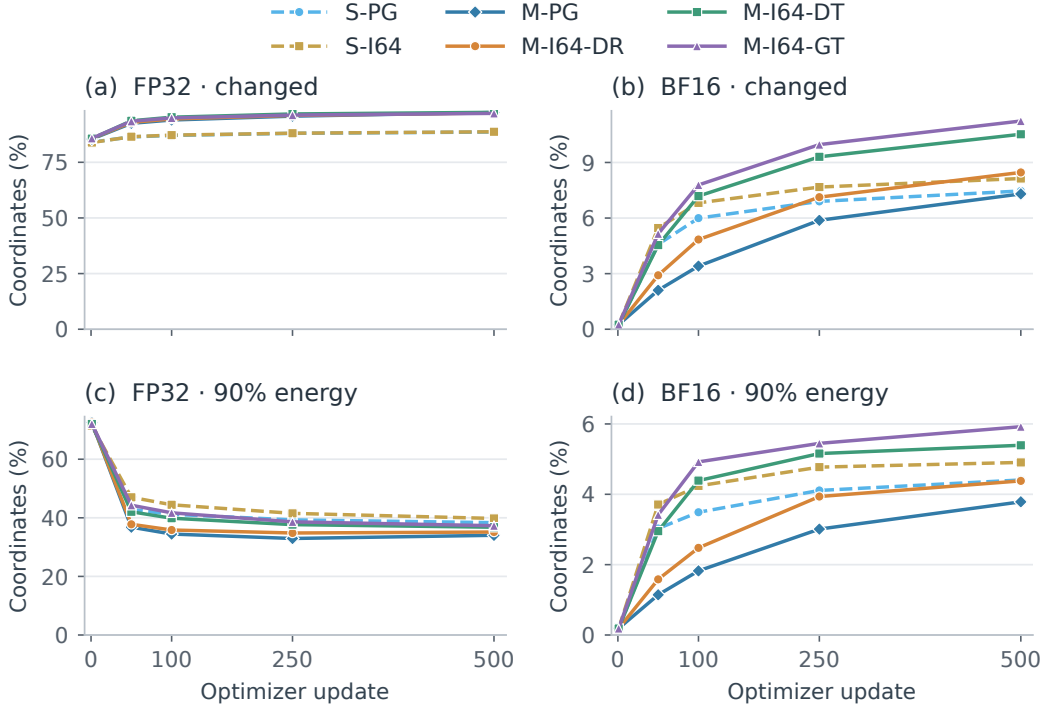


Figure 10: **Cumulative BF16 changes remain much sparser than FP32 changes.** PG/I64 trajectories under DR in single-teacher (S) and joint (M) training, with joint I64-DT/GT added: (a,b) nonzero coordinate fractions; (c,d) coordinate fractions containing 90% of squared displacement from initialization. Lines connect measured updates 1, 50, 100, 250, and 500; dashed lines denote S. No training-seed uncertainty is estimated.

response weights therefore do not imply equal gradient scale at every prefix. Measuring the retained mass and empty-intersection rate by domain can reveal unequal coverage under a fixed candidate size. Enlarging the candidate set or renormalizing its intersection changes the objective and requires a separate performance comparison.

G ADDITIONAL TEST RESULTS AND ROBUSTNESS CHECKS

G.1 ADDITIONAL TEST SETS FOR THE SAME FOUR DOMAINS

The ten joint-training configurations use the development benchmarks and checkpoint grid in Appendix A. The following analyses evaluate selected checkpoints on separate test questions, measure candidate-set coverage, and check sensitivity to known training–evaluation overlap.

Table 9 uses additional test sets fixed before inference, with 256 questions per domain: 1,024 questions and 1,792 responses per model. Mathematics and code come from the Skywork-OR1-RL-Data mathematics and TACO-based code pools; instruction following uses Nemotron-RL-instruction_following; science uses the Nemotron-RL-knowledge-mcqa validation split restricted by a fixed natural-science keyword filter. Questions are the first eligible unique candidates in the fixed source order, selected without model scores. Normalized exact-match and lexical near-duplicate checks remove questions matching the original training bundle or known evaluation pools. These checks do not establish exhaustive semantic decontamination or rule out pretraining exposure.

Scoring uses mathematics answer verification, code unit tests, instruction-following constraints, and science multiple-choice accuracy. Mathematics, code, and instruction following use greedy decoding; science uses four responses per question at temperature 1. The context budget is 32,768 tokens, with thinking disabled. The data distributions and code/instruction-following scorers differ from the

Table 9: **Method rankings depend on the task and test set.** Additional test-set scores (%) for the same four domains; mean weights domains equally.

Configuration	Update	Math	Code	IF	Science	Mean
Initial student	0	42.97	23.44	25.78	62.21	38.60
PG-DR / Adam	500	46.48	26.17	37.50	64.06	43.55
PG-DT / Adam	250	44.53	26.56	36.72	64.75	43.14
PG-DT / Adam	500	44.92	27.34	38.28	63.48	43.51
PG-GT / Adam	500	44.92	28.91	39.06	65.14	44.51
I16-DR / Adam	500	44.92	28.52	40.23	64.55	44.56
I64-DR / Adam	500	46.88	26.17	39.06	65.72	44.46
I64-DT / Adam	500	43.36	27.73	39.06	64.55	43.68
I64-GT / Adam	500	45.31	27.34	41.02	66.21	44.97
PG-DR / SGD	500	46.88	27.73	37.11	62.21	43.48
PG-DT / SGD	500	46.48	28.52	37.11	63.48	43.90
PG-GT / SGD	500	44.53	26.17	36.72	64.45	42.97

development benchmarks, so their scores are reported separately. The four-domain mean weights domains equally. Question-paired bootstrap intervals quantify finite-test uncertainty and do not estimate variation from training seeds.

G.2 CHECKPOINT SELECTION BEFORE TEST EVALUATION

We apply the three selection rules in Section 6 to the development grid in Appendix A. The grid contains updates 50/100/250/500 for seven Adam trajectories and 100/250/500 for three SGD trajectories. The retention tolerance is two percentage points per domain relative to the initial student, fixed before the additional test results. Both development rules keep the initial student unless the selected equal-domain mean improves by at least one percentage point. Checkpoint choices are fixed before test inference and remain unchanged after the test and overlap-sensitivity analyses. Both development-based rules choose update 250 for PG-DT/Adam and update 500 for the other nine trajectories. The separate PG-DT/Adam update-250 test evaluation allows direct comparison with its update-500 endpoint.

Table 2 in the main text summarizes the resulting checkpoint choices and test-score differences.

Table 10 gives all 37 trained-checkpoint development evaluations, including the ten endpoints already reported in Table 1. These intermediate measurements form the selection grid; they are not additional training runs.

The retention constraint does not change the checkpoint selected by maximum development mean for any trajectory. On the additional test sets, the update-250 PG-DT/Adam checkpoint scores 0.366 percentage points below its update-500 endpoint in four-domain mean. Thus earlier selection by development mean does not improve the separate-test mean in this case. All selected models satisfy the retention and useful-gain conditions when assessed using test point estimates, which is not a simultaneous confidence guarantee. PG-DR/Adam nevertheless has two development checkpoints outside the retention tolerance: GPQA falls from 32.20% initially to 28.03% at update 50 and 30.18% at update 250, before reaching 35.97% at update 500. Evaluating only the final checkpoint would miss these intermediate regressions.

G.3 PROBABILITY MASS RETAINED BY THE CANDIDATE INTERSECTION

For each of four student checkpoints (M-PG and M-I64-DR at updates 100 and 500), four cached response batches provide one response per domain and six sampled positions per response: 96 prefixes per checkpoint. Each response is scored by its domain teacher. We measure $p_\theta(I | h)$ for the actual intersection $I = S \cap T$ and the retained fraction $r = p_\theta(I | h) / p_\theta(S | h)$.

The minimum student probability mass in the actual student–teacher top-64 intersection is 96.55% over the 384 sampled prefixes, and no intersection is empty (Table 11). Thus the single-step I64/full-vocabulary comparisons use prefixes where truncation discards little student probability mass. They do not establish the same behavior when the intersection has low probability mass, with other teach-

Table 10: **Complete joint-training checkpoint scores.** All 37 evaluated checkpoints from ten Qwen3-1.7B joint-training configurations. Scores (%) use the original MATH-500, LiveCodeBench, IFBench strict, and GPQA avg@4 protocol; Mean is the unweighted average of the four domains. † marks the ten update-500 rows also reported in Table 1.

Configuration	Update	MATH	Code	IF	GPQA	Mean
PG-DR / Adam	50	69.60	16.41	18.67	28.03	33.18
	100	72.00	17.19	19.33	31.19	34.93
	250	73.40	16.41	26.00	30.18	36.50
	500 [†]	73.20	18.27	27.00	35.97	38.61
PG-DT / Adam	50	71.60	17.19	19.00	34.34	35.53
	100	74.60	19.53	18.00	34.97	36.78
	250	77.00	19.53	25.00	33.59	38.78
	500 [†]	73.20	19.24	27.00	34.96	38.60
PG-GT / Adam	50	71.20	18.75	19.00	34.72	35.92
	100	74.80	21.88	17.00	31.94	36.41
	250	74.80	19.53	25.00	33.84	38.29
	500 [†]	76.20	17.92	27.33	32.94	38.60
I16-DR / Adam	50	69.80	17.19	17.33	31.94	34.07
	100	72.20	17.97	19.00	32.95	35.53
	250	75.60	18.75	25.67	33.59	38.40
	500 [†]	75.40	17.75	27.00	35.46	38.90
I64-DR / Adam	50	69.60	17.19	19.00	31.69	34.37
	100	71.60	17.97	19.00	32.70	35.32
	250	74.80	17.97	26.00	33.96	38.18
	500 [†]	75.80	18.30	27.33	34.71	39.04
I64-DT / Adam	50	72.20	18.75	17.00	32.32	35.07
	100	74.80	19.53	19.33	33.33	36.75
	250	75.40	21.09	25.33	32.70	38.63
	500 [†]	74.00	18.86	28.67	35.22	39.19
I64-GT / Adam	50	74.20	17.97	17.00	32.95	35.53
	100	74.40	17.97	21.00	32.07	36.36
	250	74.40	18.75	25.00	33.33	37.87
	500 [†]	74.20	19.80	28.67	33.41	39.02
PG-DR / SGD	100	73.00	18.75	25.67	35.35	38.19
	250	75.80	18.75	25.67	31.82	38.01
	500 [†]	76.40	19.53	28.33	35.59	39.96
PG-DT / SGD	100	74.80	18.75	23.33	34.34	37.81
	250	74.20	19.53	25.33	35.10	38.54
	500 [†]	75.20	20.31	28.33	33.90	39.44
PG-GT / SGD	100	74.60	17.97	20.67	35.73	37.24
	250	73.00	20.31	23.67	33.33	37.58
	500 [†]	75.00	19.26	25.33	34.71	38.58

Table 11: **The minimum observed I64 intersection mass is 96.55%.** Statistics over sampled prefixes; $p(I)$ is in percent and $r = p_\theta(I | h)/p_\theta(S | h)$.

Checkpoint	$p(I)$: mean	$p(I)$: min	$ I $: mean	$ I $: min	r : mean	r : min
M-PG/100	99.9141	96.5482	58.96	43	0.9999413	0.9977714
M-PG/500	99.9952	99.7825	61.59	50	0.9999975	0.9998511
M-I64-DR/100	99.9564	97.5664	60.08	36	0.9997331	0.9757993
M-I64-DR/500	99.9826	99.3722	61.75	51	0.9999912	0.9997005

Table 12: **Removing four overlapping mathematics questions can reorder near-tied configurations.** Mean (496) substitutes MATH-496 for MATH-500 and keeps the other domain scores fixed.

Configuration	Update	MATH-500	MATH-496	Mean (500)	Mean (496)
Initial student	0	68.80	69.15	33.94	34.02
PG-DR / Adam	500	73.20	73.79	38.61	38.76
PG-DT / Adam	250	77.00	77.42	38.78	38.89
PG-DT / Adam	500	73.20	73.59	38.60	38.70
PG-GT / Adam	500	76.20	76.81	38.60	38.75
I16-DR / Adam	500	75.40	76.01	38.90	39.06
I64-DR / Adam	500	75.80	76.41	39.04	39.19
I64-DT / Adam	500	74.00	74.40	39.19	39.29
I64-GT / Adam	500	74.20	74.60	39.02	39.12
PG-DR / SGD	500	76.40	77.02	39.96	40.12
PG-DT / SGD	500	75.20	75.81	39.44	39.59
PG-GT / SGD	500	75.00	75.60	38.58	38.73

ers, or during I16 training. The four batches repeat response sampling at each checkpoint; they are not independent training seeds, and responses may differ between checkpoints.

G.4 SENSITIVITY TO KNOWN MATHEMATICS OVERLAP

The fixed Qwen training pool contains five records matching four known MATH-500 problems, including variants in wording or answer format. The affected MATH test identifiers are counting-and-probability problem 430 and intermediate-algebra problems 1849, 2146, and 582. Their presence in the pool does not show whether a run sampled them or whether they improved its score. We remove these four questions from the existing per-question evaluation results, retaining the original 500-question results as the main protocol.

Removing the four questions changes the four-domain means by approximately 0.09–0.16 points and can reorder near-tied configurations. PG-DT/Adam and PG-GT/Adam both round to 38.60% before removal, but reach 38.70% and 38.75%, respectively, afterward. PG-DR/SGD retains the highest mean among the reported checkpoints. MATH-496 names the retained subset, not a new official benchmark. We keep the checkpoint choices fixed. This analysis does not retrain on a filtered pool or provide an exhaustive contamination audit.

G.5 TEACHER-UPDATE OVERLAP AND DISTRIBUTIONAL SIMILARITY

Figure 11 supplements the zero-gradient baseline subtraction at M-PG/500 using the controls in Appendix E. It compares PG and I64 under Adam with trained moments, reset- m , and norm-matched SGD. Shared inputs hold student prefixes fixed between teachers; domain-specific inputs change both the teacher and the input domain. Each of four batches contains six teacher pairs; these observations are dependent because they reuse teachers and responses. These gradient and update measurements do not directly measure changes in task performance.

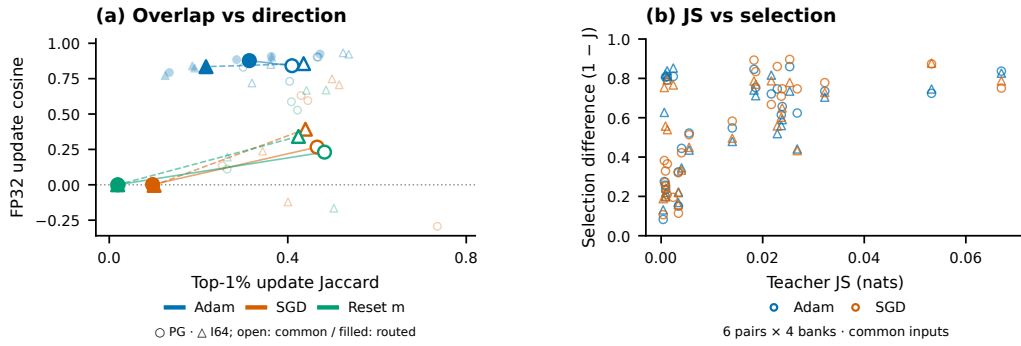


Figure 11: **Coordinate overlap and update direction measure different properties.** (a) Top-1% FP32 update-set Jaccard versus cosine; small/large points show batch/overall means. (b) Teacher-distribution JS divergence versus $1 - \text{Jaccard}$ on shared inputs; each point is a teacher pair within a batch.

G.6 ONLINE COVERAGE AND PER-DOMAIN NORMALIZATION EFFECTS

Online I16 coverage. For I16-DR, each of 500 updates has 16 logged slices of four responses. We average their masked, response-averaged summaries equally per update, rather than pooling tokens. Candidate sets are fixed at rollout; student probabilities are measured during training. Across 8,000 slices, mean $p_\theta(I)$, r , and $|I|$ are 99.616%, 99.879%, and 15.048. The mean empty fraction is 0.00270%, nonzero in 80 slices across 62 updates. The lowest slice-mean mass is 86.67% and largest slice-mean empty fraction 1.85%; these are not prefix-level extremes. Different inputs and aggregation preclude a paired comparison with the offline I64 measurements (Table 11).

Per-domain probe losses. We disaggregate Figure 2(c), using the same six batches and twelve branches per batch (Appendix B.1). Each batch has one held-out response per domain. At update 250, saved-state Adam under DR/DT/GT increases IF KL by $0.362/0.293/0.272 \times 10^{-3}$, while the four-domain changes are $0.041/-0.002/-0.027 \times 10^{-3}$. Thus DT/GT lower the mean despite an IF increase. These local loss changes do not establish long-term capability regressions.

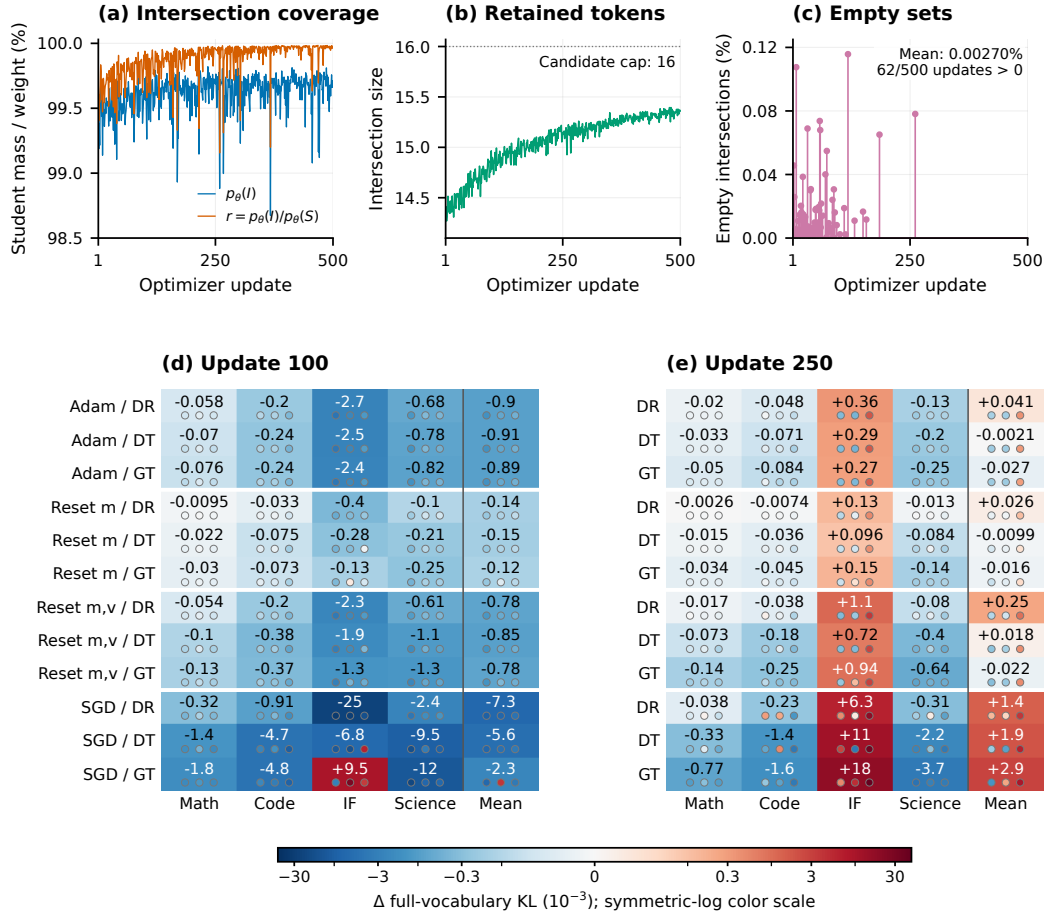


Figure 12: **High average coverage and low mean probe loss can conceal exceptions.** (a–c) I16 per-update means of 16 logged slices, without temporal smoothing; the mass axis spans 98.5–100%. (d,e) Full-vocabulary KL after minus before BF16 writeback, in units of 10^{-3} . Cells show three-batch means; dots show batches 1042–1044 from left to right. The shared symmetric-log color scale is linear within ± 0.1 . SGD is matched to each rule’s Adam step norm.

H SMOLLM3 DIAGNOSTICS AND CAPABILITY EVALUATION

H.1 CHECKPOINTS, RESPONSES, AND WEIGHTING

We use the SmolLM3-3B MixSFT student, three domain-specific RL teachers, and exported 500-update students for I64-DR/DT/GT and PG-DR/SGD. The model and data revisions, export hashes, and diagnostic source snapshots are recorded with the measurements. The reported I64 training configurations use Adam with learning rate 2.5×10^{-7} ; PG uses SGD with learning rate 0.0025. We do not use historical aggregate capability scores as a matched evaluation, and the three original I64 training source revisions and prompt schedules were not fully recovered. The exported checkpoints do not establish that those runs differed only in loss reduction.

Diagnostic prompts come from the frozen Open-MOPD data revision 9e897efe, with unique raw messages shuffled within mathematics, code, and instruction following. Coverage, normalization, and probe prompt splits are disjoint within this diagnostic sample; absence from the original training schedule is not established. We preserve dataset messages and disable thinking in the chat template. Generation uses temperature 1, top- $p = 1$, a 2,048-token prompt cap, and a 4,096-token response cap. Each natural coverage bank contains 384 responses, 128 per domain, with up to eight uniformly spaced positions per response: 3,053 prefixes for MixSFT and 3,058 for DR500. Additional students score the same 3,058 DR500 prefixes without regenerating responses.

Coverage statistics assign equal weight to domains, responses within domains, and audited positions within a response. Mean-mass intervals resample responses independently within each domain 2,000 times using seed 1042/1043/1044. They do not estimate training-seed variation. The first two rows of Table 13 use different natural banks; the remaining students use the DR500 bank. The minimum I64 mass on DR500’s natural bank is 1.7454×10^{-8} ; DT500 and GT500 each have one empty intersection in the common-bank audit. Means therefore do not imply uniformly high coverage.

Table 13: **High average I64 coverage coexists with a lower tail.** SmolLM3, 384 responses per bank. Mass and frequency columns are percentages; intervals resample responses within domains. The last column counts empty intersections among audited prefixes.

Student	Response bank	Mean mass [95% CI]	< 90%	1st pct.	Empty
MixSFT	MixSFT	97.42 [96.93, 97.88]	8.42	62.06	0
DR500	DR500	98.56 [98.16, 98.96]	3.65	64.75	0
GT500	DR500	98.54 [98.13, 98.94]	3.61	64.79	1
DT500	DR500	98.58 [98.18, 98.97]	3.52	65.41	1
PGSGD500	DR500	98.65 [98.27, 99.03]	3.52	66.49	0

H.2 FROZEN SUBSETS AND GRADIENT COMPARISONS

Within each domain we select representative prefixes, prefixes above its 75th entropy percentile, and prefixes with I64 mass below 0.99. Selection precedes the objective-gradient comparison, allows overlap between strata, and uses up to 32 distinct prompts per domain. The low-coverage code stratum has only 21 eligible prompts, giving 85 prefixes overall rather than 96; no duplicated positions fill the gap. Each stratum is divided into three batches. Small candidate budgets $k = 4, 8$ are selected by a coverage-only rule before inspecting objective gradients.

The BF16 coverage audit sorts a top-128 list and takes its first k entries, whereas gradient diagnostics recompute top- k directly in FP32 at the exported BF16 parameter values. Student forward/backward passes use FP32; cached teacher targets use BF16 forward passes followed by FP32 log-softmax. Thus their probabilities and threshold membership need not match bit for bit; one of the 85 frozen low-coverage prefixes crosses 0.99 during the FP32 diagnostic, and remains in its original stratum. The FP32 diagnostic mean I64 masses are 99.049%, 95.684%, and 93.121% for the representative, high-entropy, and low-coverage groups. These are domain-weighted summaries of the selected prefixes, not estimates of their prevalence in natural generation.

Full KL, PG, and intersection objectives use the definitions in Appendix D. Empty intersections produce zero loss. We retain zero- and small-norm cases and report vector differences in addition

to cosine. The plotted error is $\|g_{I_k} - g_{full}\|/\|g_{full}\|$, not the difference of the two gradient norms. Batch means and thin batch trajectories in Figure 6 make the diagnostic replication unit explicit.

H.3 OPTIMIZER STATE, SAMPLING, AND NUMERICAL PRECISION

Normalization uses three separate batches, each with eight responses per domain, and the true response lengths for GT/DT weights. Raw gradients, fresh Adam, and SGD start from independent copies of the same exported parameters. Fresh Adam starts with both moments zero and performs its first step; SGD uses learning rate 0.0025, zero momentum, and norm-1 clipping. Fresh Adam is distinct from the Qwen reset- m control, which retains the trained second moment and step counter. We do not match their norms to an unavailable native Adam step.

For all three rule pairs in all three batches, fresh Adam lowers cosine relative to the raw gradients. For example, DT-DR raw cosines are 0.8968/0.8655/0.5933, versus 0.7013/0.6489/0.4126 after fresh Adam. This qualifies an optimizer-wide interpretation of alignment: the effect depends on state, and the native SmolLM3 history remains untested.

Conditional PG variance uses only three fixed prefixes per group, one per domain, with eight independent action-sampling repetitions. Sample budgets $m = 1, 16, 64$ use nested actions within a repetition. We normalize the total coordinate sample variance by $\|g_{full}\|^2$ and separately report the error of the eight-repetition mean. The representative-group variance is 1.426/0.570/0.123; the low-coverage-group variance is 4.965/2.566/0.434. These finite conditional measurements do not establish an exact $1/m$ law or training-run variance.

The precision controls begin exactly at parameter values represented by BF16, cast to FP32 for computation. Their BF16 writeback depends on this initial position on the quantization grid, so single-step changed fractions need not reproduce native training with accumulated rounding residuals. Native DR500 FP32 master weights, Adam moments and step, and the initialization master were not available for these diagnostics. We therefore report no native-history/reset- m comparison or cumulative FP32 change for SmolLM3.

H.4 CUMULATIVE BF16 CHANGES BY PARAMETER GROUP

The exported MixSFT-to-DR500 comparison contains 3,075,098,624 unique parameter coordinates after accounting for tied embeddings. The 36 transformer blocks each contain 78,123,008 coordinates, the token embedding 262,668,288, and the final normalization 2,048. We compare the two BF16 exports directly. Overall, 10.4695% of coordinates change; the largest 1% carry 36.8477% of squared displacement, and 5.8496% of coordinates carry 90%. Figure 14 retains all parameter groups. Per-group changed fractions use each group’s own coordinate count; displacement shares use total model squared displacement. Neither a local nor a cumulative BF16 statistic supplies the missing FP32 history.

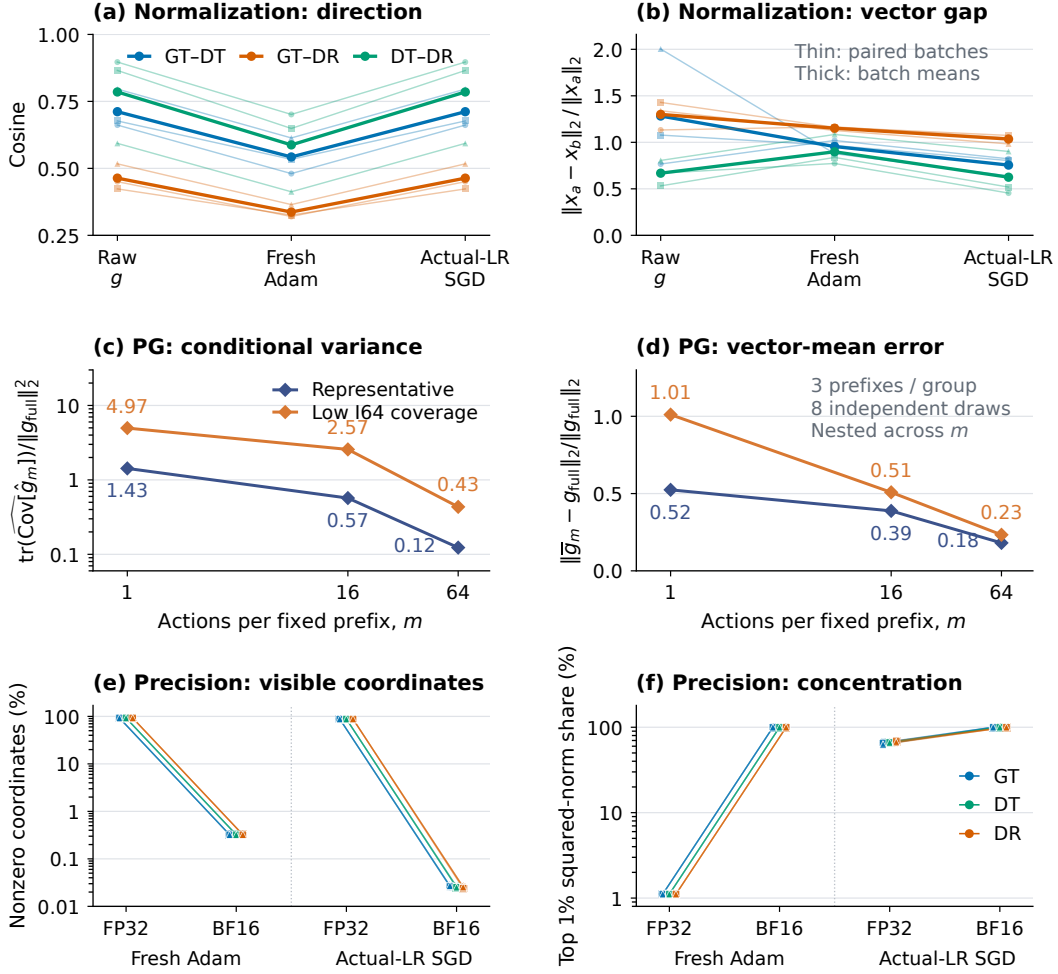


Figure 13: **Local controls separate optimizer, sampling, and precision effects.** (a,b) Three paired SmolLM3 batches (thin) and their means (bold). In (b), x is the gradient or step, normalized by the first named rule. (c,d) Three fixed prefixes per group, eight action repeats; (d) error of the vector mean. (e,f) Each rule/batch pairs FP32 changes with BF16 writeback. Fresh Adam uses zero moments at exported BF16 parameters, not native master weights or trained history.

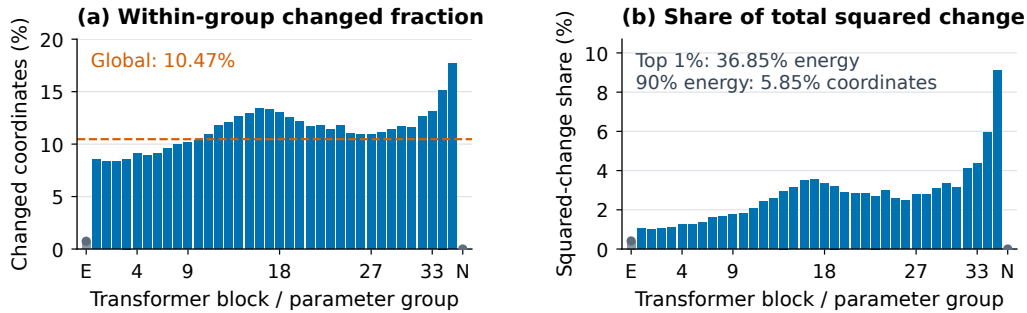


Figure 14: **Cumulative BF16 changes differ across all 38 SmolLM3 parameter groups.** MixSFT to DR500: blocks 0–35, embedding (E), final norm (N). (a) Within-group changed fraction; dashed: global parameter-weighted fraction. (b) Share of total squared displacement. Tied weights count once; native FP32 changes are unavailable.

H.5 ENDPOINT CAPABILITY EVALUATION

Table 14 evaluates the exported students with shared prompts, scorers, and decoding settings within each benchmark. We use BF16, disable thinking, and allocate a 32,768-token context with the response budget equal to context length minus prompt length. The SmolLM evaluation uses the separately recorded 128-question LiveCodeBench subset; the main Qwen study uses 1,055 questions. MATH-500, this LiveCodeBench subset, and IFBench use greedy decoding; GPQA uses temperature 1 and top- $p = 1$; AIME uses temperature 0.6 and top- $p = 0.95$. Budget-capped outputs remain in scoring. Score and paired-difference intervals use 100,000 question-bootstrap replicates (seed 1042/1043/1044), retaining all responses to each sampled question.

Table 14: **SmolLM3 capability under a shared evaluation protocol.** Scores (%): AIME pass@8, GPQA average over four responses, otherwise greedy accuracy. $Q \times R$: questions $\times$ responses; Resp.: average response accuracy; Tokens: mean generated length; Cap: budget-capped responses (%). Intervals resample questions.

Benchmark	$Q \times R$	Score [95% CI]	Resp.	Tokens	Cap (%)
<i>MixSFT initialization</i>					
MATH-500 [†]	500×1	50.60 [46.20, 55.00]	50.60	1501	3.00
AIME 2024	30×8	10.00 [0.00, 23.33]	2.92	3840	5.83
AIME 2025	30×8	13.33 [3.33, 26.67]	1.67	2735	3.75
<i>I64-DT/Adam, update 500</i>					
AIME 2024	30×8	26.67 [13.33, 43.33]	9.58	5019	7.50
AIME 2025	30×8	13.33 [3.33, 26.67]	3.75	1227	0.00
<i>PG-DR/SGD, update 500</i>					
MATH-500	500×1	53.80 [49.40, 58.20]	53.80	1520	2.80
LiveCodeBench v6	1055×1	17.97 [11.72, 25.00]	17.97	1920	5.47
IFBench (strict)	300×1	20.33 [16.00, 25.00]	20.33	2066	5.33
GPQA Diamond	198×4	23.99 [20.71, 27.40]	23.99	543	0.00
AIME 2024	30×8	16.67 [3.33, 30.00]	4.17	2215	1.67
AIME 2025	30×8	13.33 [3.33, 26.67]	2.08	4957	9.17

[†] Additional complete MixSFT run under the same protocol (500 scored responses).

Relative to MixSFT, PG/SGD changes MATH-500 accuracy by +3.20 points (95% paired interval $[-0.20, 6.60]$), gaining 45 questions and losing 29. I64-DT changes AIME 2024 pass@8 by +16.67 points $[0.00, 33.33]$. On AIME 2025, MixSFT and PG/SGD share only one of their four solved questions despite equal pass@8 (Figure 15).

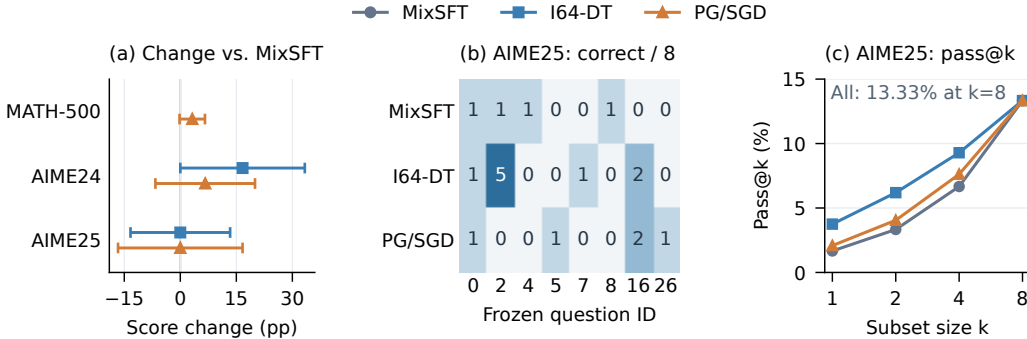


Figure 15: **Equal aggregate scores can conceal different solved questions and sampling outcomes.** (a) Differences from MixSFT with paired question-bootstrap 95% intervals: accuracy for MATH-500, pass@8 for AIME. (b) Correct responses out of eight for every AIME 2025 question solved by any model; the other 22 questions have zero successes. (c) Subset estimates from the same eight responses: $\widehat{\text{pass@}k} = \frac{1}{30} \sum_q [1 - \binom{8-c_q}{k} / \binom{8}{k}]$, where c_q counts correct responses.